Fanchen Bu

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Education

Korea Advanced Institute of Science and Technology (KAIST), South Korea 2022.03 – Present
Ph.D. student in Electrical Engineering (expected Spring 2027)
Supervisor: Prof. Kijung Shin

Korea Advanced Institute of Science and Technology (KAIST), South Korea 2019.09 – 2021.08
M.S. in Electrical Engineering
Thesis: A Novel Optimization Algorithm with Orthogonality for Deep Neural Networks Inspired by Feedback Integrators
Supervisor: Prof. Dong Eui Chang

University of Chinese Academy of Sciences (UCAS), China 2015.09 – 2019.08
B.Eng. in Computer Science and Technology
Thesis: Vehicle Trajectory Prediction Based on Deep Learning
Supervisor: Prof. Dongbin Zhao

Experience

KTH Royal Institute of Technology 2026.09 – 2026.11
Visiting Researcher Host: Prof. Aristides Gionis

CENTAI Institute (now Intesa Sanpaolo AI Research), Italy 2025.03 – 2025.06
Visiting Researcher Host: Dr. Francesco Bonchi | Collaborator: Dr. Atsushi Miyauchi

Korea Advanced Institute of Science and Technology (KAIST), South Korea 2021.09 – 2022.02
Research assistant Supervisor: Prof. Kijung Shin

Publications **Google Scholar**

(C: Conference / J: Journal / P: Preprint / W: Workshop / *: Equal contribution)

[P2] Fanchen Bu, Fan Li, Geon Lee, Sunwoo Kim, Xiaoyang Wang, Renaud Lambiotte, and Kijung Shin. "Do Higher-Order Models Win for Higher-Order Reasons? Rethinking Performance Gains in Hypergraph Learning." *arXiv:2610.07981.*

[P1] Soo Yong Lee*, Jongha Lee*, Jaewan Chun, Hyunjin Hwang, Fanchen Bu, Ziv Ben-Zion, Taekwan Kim, Denny Borsboom, Jaemin Yoo, and Kijung Shin. "Automating and Scaling Behavioral Scientific Research on AI Agents." *arXiv:2608.10030.*

[J6] Jaewan Chun*, Fanchen Bu*, Yeongho Kim, Atsushi Miyauchi, Francesco Bonchi, and Kijung Shin. "A Survey on Centrality and Importance Measures in Hypergraphs: Categorization and Empirical Insights." *ACM Computing Surveys (CSUR) 2026.*

[J5] Geon Lee, Kyungho Kim, Fanchen Bu, Langzhang Liang, and Kijung Shin. "Revisiting LightGCN: Unexpected Inflexibility, Inconsistency, and A Remedy Towards Improved Recommendation." *ACM Transactions on Recommender Systems (TORS) 2026.*

[C21] Geon Lee*, Fanchen Bu*, Soo Yong Lee*, Sunwoo Kim, and Kijung Shin. "Toward a Science of AI Agent Societies." *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2026. (Blue Sky Ideas Track)*

[C20] Fanchen Bu, Ruochen Yang, Paul Bogdan, and Kijung Shin. "Edge Probability Graph Models Beyond Edge Independency: Concepts, Analyses, and Algorithms." *IEEE International Conference on Data Mining (ICDM) 2025.*

[C19] Fanchen Bu, Geon Lee, Minyoung Choe, and Kijung Shin. "Identifying Group Anchors in Real-World Group Interactions Under Label Scarcity." *IEEE International Conference on Data Mining (ICDM) 2025. (Short paper)*

[C18] Hyunjin Choo, Fanchen Bu, Hyunjin Hwang, Young-Gyu Yoon, and Kijung Shin. "HyperSearch: Prediction of New Hyperedges through Unconstrained yet Efficient Search." *IEEE International Conference on Data Mining (ICDM) 2025.*

[C17] Fanchen Bu and Kijung Shin. "PyTorch-based Geometric Learning with Non-CUDA Processing Units: Experiences from Intel Gaudi-v2 HPUs." *Korea Computer Congress (KCC) 2025.*

[C16] Federico Berto*, Chuanbo Hua*, Junyoung Park*, Laurin Luttmann*, Yining Ma, Fanchen Bu, Jiarui Wang, Haoran Ye, and 22 Other Authors. "RL4CO: an Extensive Reinforcement Learning for Combinatorial Optimization Benchmark." *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2025. (Oral)*

[C15] Langzhang Liang, Fanchen Bu, Zixing Song, Zenglin Xu, Shirui Pan, and Kijung Shin. "Mitigating Over-Squashing in Graph Neural Networks by Spectrum-Preserving Sparsification." *International Conference on Machine Learning (ICML) 2025.*

[C14] Hyeonsoo Jo, Jongha Lee, Fanchen Bu, and Kijung Shin. "TiGer: Self-Supervised Purification for Time-evolving Graphs." *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD) 2025.*

[C13] Junghun Lee, Hyunju Kim, Fanchen Bu, Jihoon Ko, Kijung Shin. "DiffIM: Differentiable Influence Minimization with Surrogate Modeling and Continuous Relaxation." *AAAI Conference on Artificial Intelligence (AAAI) 2025.*

[C12] Hyeonsoo Jo*, Hyunjin Hwang*, Fanchen Bu, Soo Yong Lee, Chanyoung Park, and Kijung Shin. "On Measuring Unnoticeability of Graph Adversarial Attacks: Observations, New Measure, and Applications." *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2025.*

[J4] Geon Lee*, Fanchen Bu*, Tina Eliassi-Rad, and Kijung Shin. "A Survey on Hypergraph Mining: Patterns, Tools, and Generators." *ACM Computing Surveys (CSUR) 2025.*

[J3] Hyunju Kim*, Heechan Moon*, Fanchen Bu, Jihoon Ko, and Kijung Shin. "Estimating Simplet Counts via Sampling." *The VLDB Journal 2025.*

[W1] Fanchen Bu and Kijung Shin. "On Training-Test (Mis)alignment in Unsupervised Combinatorial Optimization: Observation, Empirical Exploration, and Analysis." *Workshop on Test-Time Adaptation @ International Conference on Machine Learning (ICML) 2025.*

[C11] Sunwoo Kim, Soo Yong Lee, Fanchen Bu, Shinhwan Kang, Kyungho Kim, Jaemin Yoo, and Kijung Shin. "Rethinking Reconstruction-based Graph-Level Anomaly Detection: Limitations and a Simple Remedy." *Conference on Neural Information Processing Systems (NeurIPS) 2024.*

[C10] Fanchen Bu, Hyeonsoo Jo, Soo Yong Lee, Sungsoo Ahn, and Kijung Shin. "Tackling Prevalent Conditions in Unsupervised Combinatorial Optimization: Cardinality, Minimum, Covering, and More." *International Conference on Machine Learning (ICML) 2024.*

[C9] Soo Yong Lee, Sunwoo Kim, Fanchen Bu, Jaemin Yoo, Jiliang Tang, and Kijung Shin. "Feature Distribution on Graph Topology Mediates the Effect of Graph Convolution: Homophily Perspective." *International Conference on Machine Learning (ICML) 2024.*

[C8] Sunwoo Kim, Shinhwan Kang, Fanchen Bu, Soo Yong Lee, Jaemin Yoo, and Kijung Shin. "HypeBoy: Generative Self-Supervised Representation Learning on Hypergraphs." *International Conference on Learning Representations (ICLR) 2024.*

[C7] Hyeonsoo Jo, Fanchen Bu, and Kijung Shin. "Robust Graph Clustering via Meta Weighting for Noisy Graphs." *ACM International Conference on Information and Knowledge Management (CIKM) 2023.*

[C6] Fanchen Bu and Kijung Shin. "On Improving the Cohesiveness of Graphs by Merging Nodes: Formulation, Analysis, and Algorithms." *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2023.*

[C5] Sunwoo Kim, Fanchen Bu, Minyoung Choe, Jaemin Yoo, and Kijung Shin. "How Transitive Are Real-World Group Interactions? - Measurement and Reproduction." *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2023.*

[C4] Soo Yong Lee, Fanchen Bu, Jaemin Yoo, and Kijung Shin. "Towards Deep Attention in Graph Neural Networks: Problems and Remedies." *International Conference on Machine Learning (ICML) 2023.*

[C3] Hyunju Kim, Jihoon Ko, Fanchen Bu, and Kijung Shin. "Characterization of Simplicial Complexes by Counting Simplets Beyond Four Nodes." *ACM Web Conference (WWW) 2023.*

[J2] Fanchen Bu, Shinhwan Kang, and Kijung Shin. "Interplay between Topology and Edge Weights in Real-World Graphs: Concepts, Patterns, and an Algorithm." *Data Mining and Knowledge Discovery (DAMI) 2023.*

[J1] Fanchen Bu, Geon Lee, and Kijung Shin. "Hypercore Decomposition for Non-Fragile Hyperedges: Concepts, Algorithms, Observations, and Applications." *Data Mining and Knowledge Discovery (DAMI) 2023.*

[C2] Fanchen Bu and Dong Eui Chang. "Feedback Gradient Descent: Efficient and Stable Optimization with Orthogonality for DNNs." *AAAI Conference on Artificial Intelligence (AAAI) 2022.*

[C1] Fanchen Bu and Dong Eui Chang. "Double Prioritized State Recycled Experience Replay." *IEEE International Conference on Consumer Electronics – Asia (ICCE-Asia) 2020.*

✎ Academic Services

(C: Conference / J: Journal)

[C11] International Conference on Learning Representations (ICLR):

- Reviewer: 2026 – 2027

[C10] AAAI Conference on Artificial Intelligence (AAAI):

- Reviewer: 2026 – 2027

[C9] ACM International Conference on Information and Knowledge Management (CIKM):

- Reviewer: 2025 – 2026
- Session Chair: 2025

[C8] Conference on Neural Information Processing Systems (NeurIPS):

- Reviewer: 2025 – 2026

[C7] The International Conference on Machine Learning (ICML):

- Reviewer: 2025 – 2026

[C6] Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD):

- Workshop organizer: 2025

[C5] ACM Web Conference (WWW):

- Reviewer: 2025 – 2027

[C4] Learning on Graphs Conference (LoG):

- Reviewer: 2024, 2026

[C3] European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD):

- Reviewer: 2024

[C2] Asian Conference on Machine Learning (ACML):

- Reviewer: 2024 – 2026

[C1] ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD):

- Reviewer: 2024 – 2027

[J16] Progress in Artificial Intelligence:

- Reviewer: 2026

[J15] Transactions on Machine Learning Research (TMLR):

- Reviewer: 2026

[J14] Data Mining and Knowledge Discovery (DAMI):

  - Reviewer: 2026

[J13] Machine Learning:

  - Reviewer: 2026

[J12] IEEE Transactions on Knowledge and Data Engineering (TKDE):

  - Reviewer: 2026

[J11] Journal of Network and Systems Management:

  - Reviewer: 2026

[J10] PLoS One:

  - Reviewer: 2026

[J9] Neural Processing Letters:

  - Reviewer: 2025

[J8] The VLDB Journal:

  - Reviewer: 2025 – 2026

[J7] Computational and Mathematical Organization Theory:

  - Reviewer: 2025

[J6] Scientific Reports:

  - Reviewer: 2025

[J5] IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI):

  - Reviewer: 2025

[J4] International Journal of Machine Learning and Cybernetics (IJMLC):

  - Reviewer: 2024

[J3] IEEE Transactions on Network Science and Engineering (TNSE):

  - Reviewer: 2024, 2026

[J2] The Journal of Supercomputing:

  - Reviewer: 2024 – 2025

[J1] Big Data Research:

  - Reviewer: 2024

♡ Honors and Awards

One of the best reviewers (top 407) in KDD'26 (both August and February Cycles)	2026
One of the top reviewers (top 200) in ICLR'26	2026
One of the Gold reviewers (top 25%) in ICML'26	2026
One of the outstanding reviewers (top 10%) in KDD'25 (both August and February Cycles)	2025
One of the top reviewers (top 32) in LoG'24	2024

Languages

- Chinese: Native
  - Native: Mandarin and Wu Chinese
  - Intermediate: Cantonese
  - Elementary: Hokkien
- English: Advanced
  - IELTS 7.5
  - GRE 333

- Korean: Intermediate
- Italian: Elementary